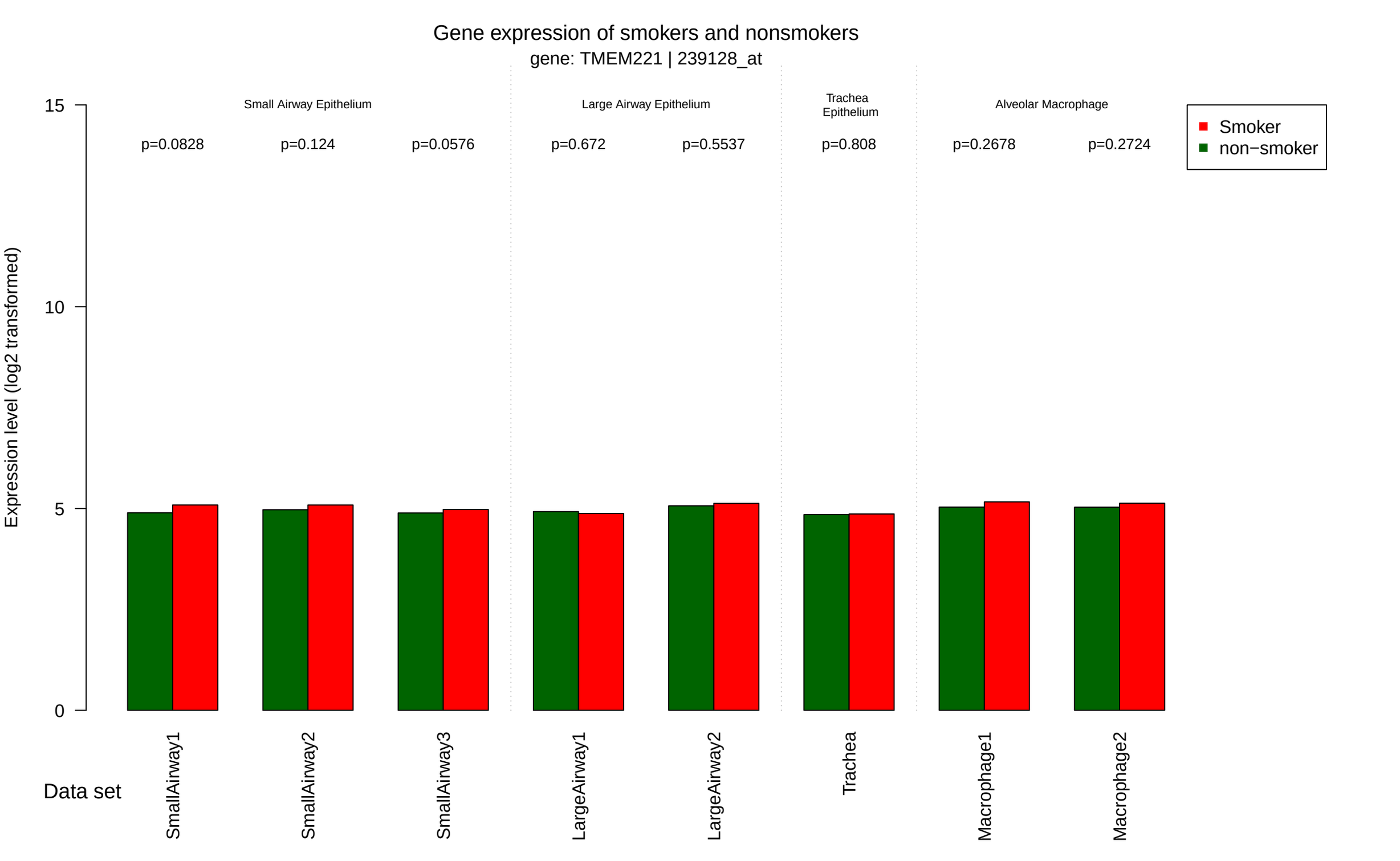
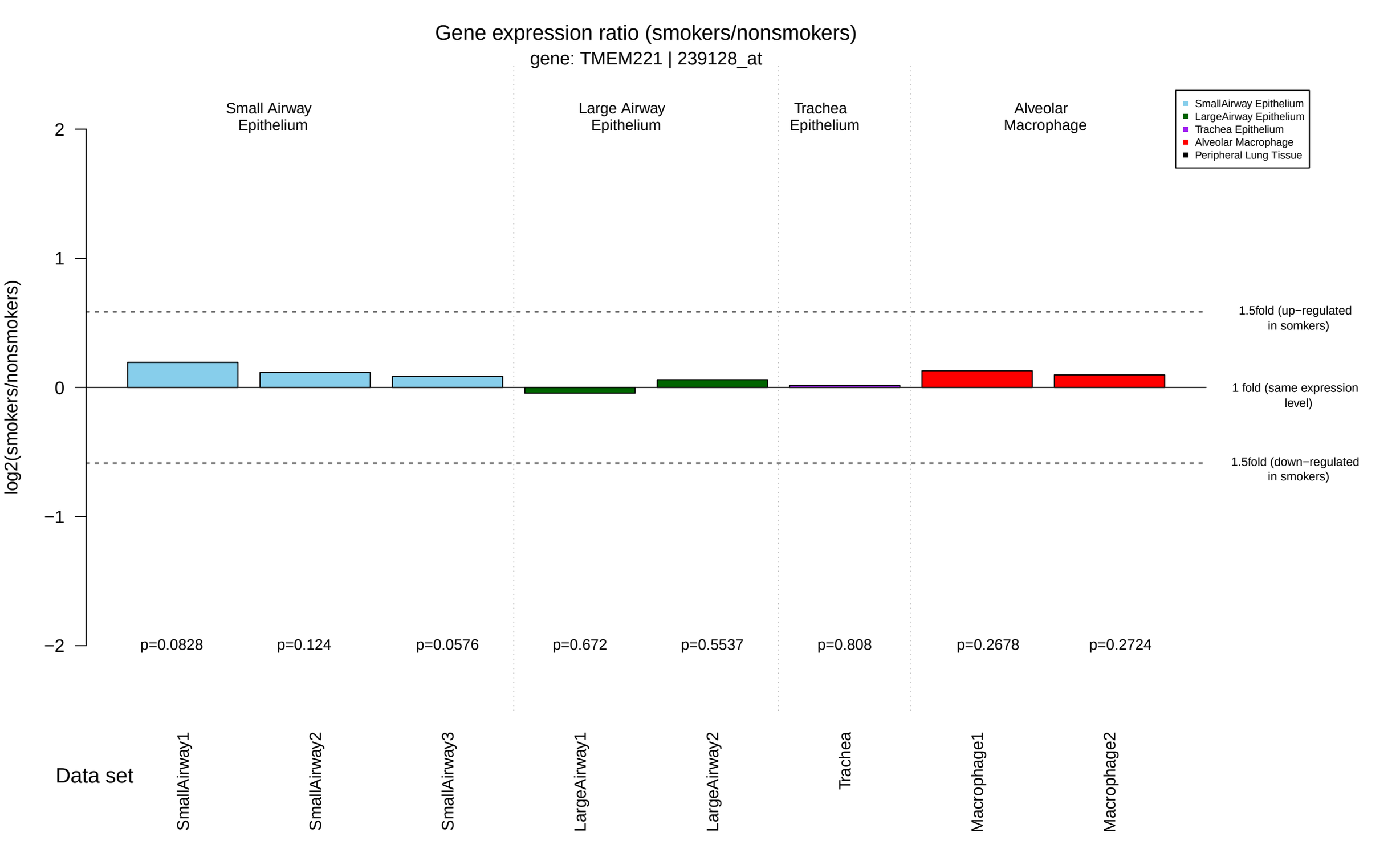
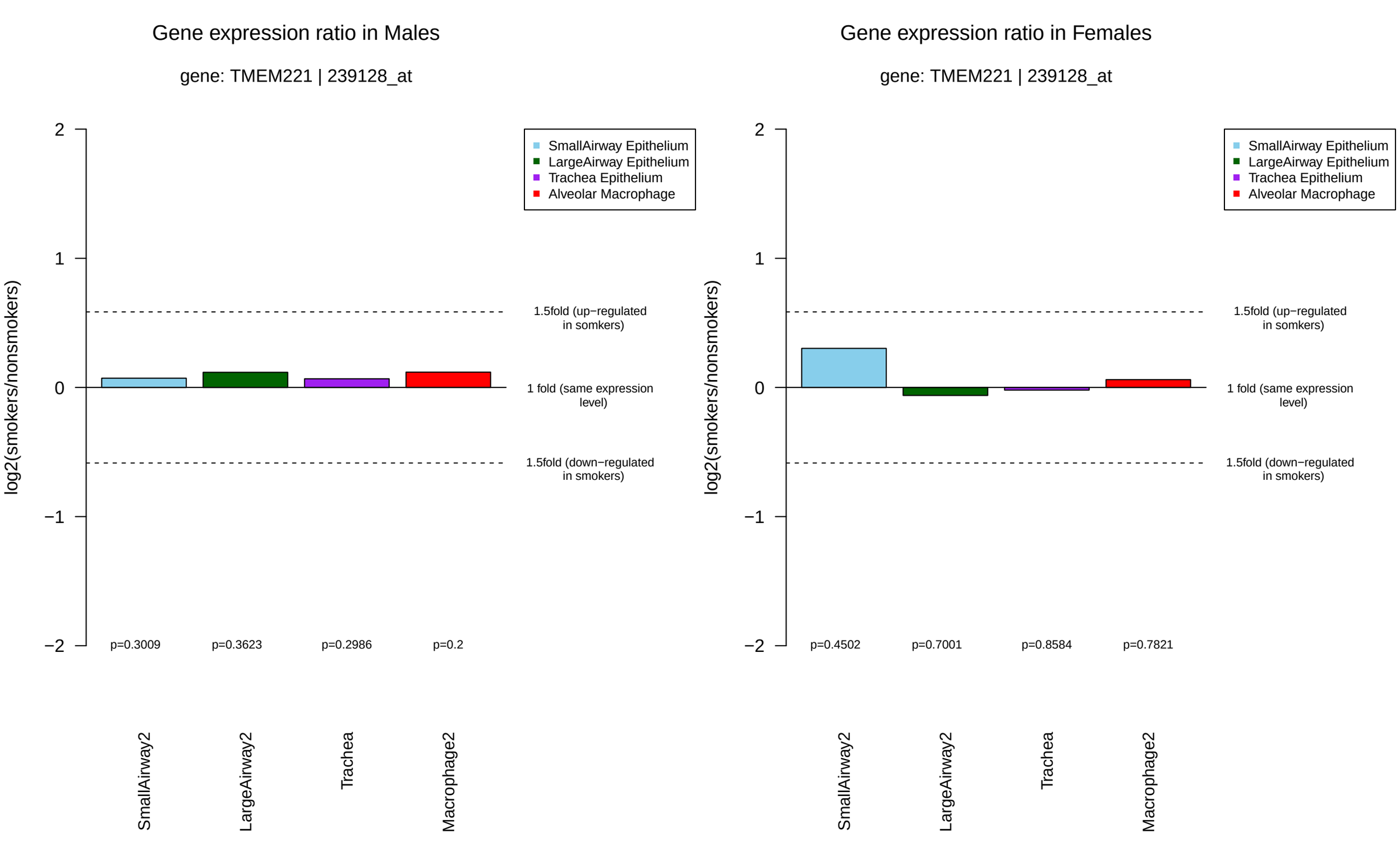
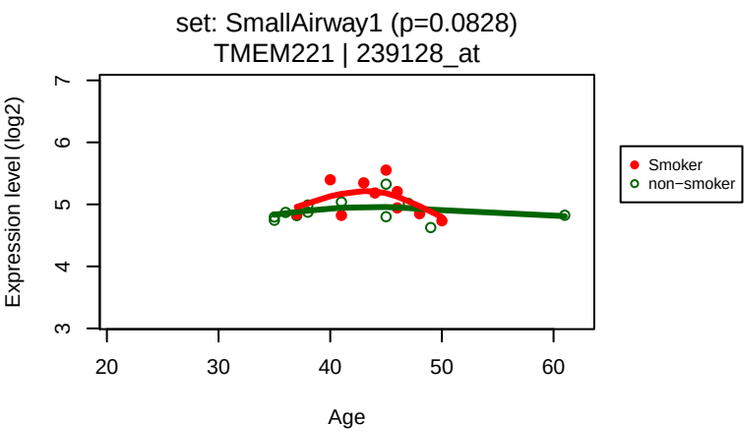
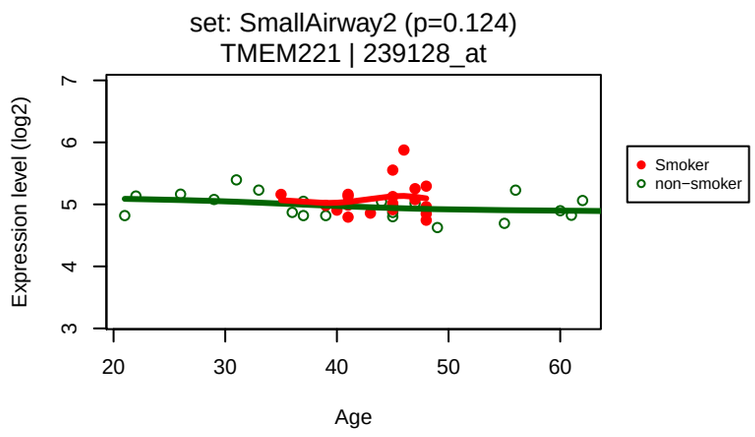
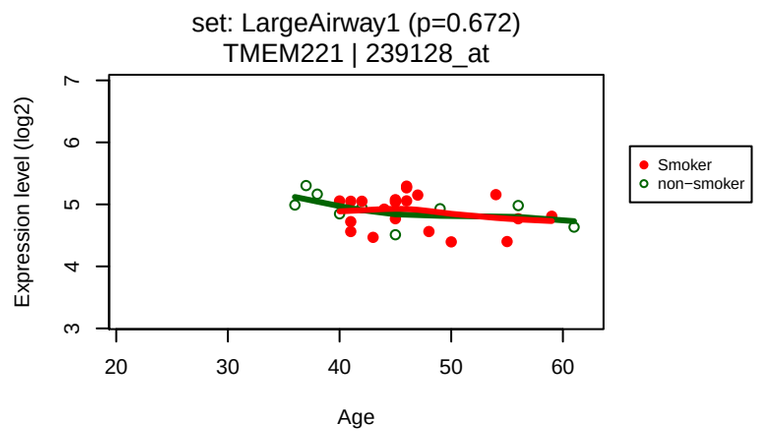

Gene expression of smokers and nonsmokers
gene: TMEM221 | 239128_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0828
p=0.124
p=0.0576
p=0.672
p=0.5537
p=0.808
p=0.2678
p=0.2724
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

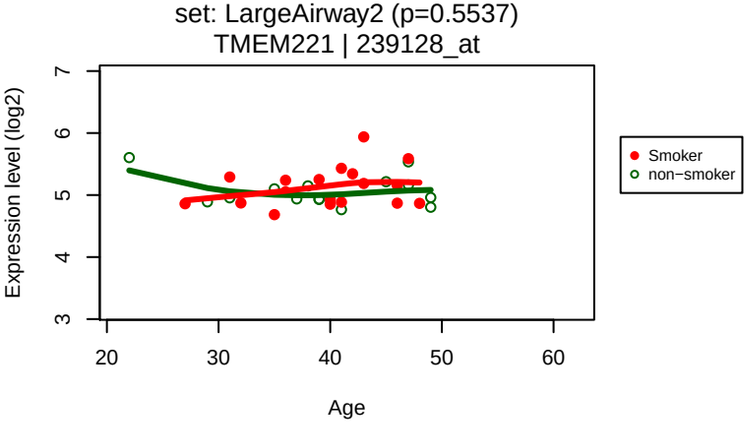

Gene expression ratio (smokers/nonsmokers)
gene: TMEM221 | 239128_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0828
p=0.124
p=0.0576
p=0.672
p=0.5537
p=0.808
p=0.2678
p=0.2724
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

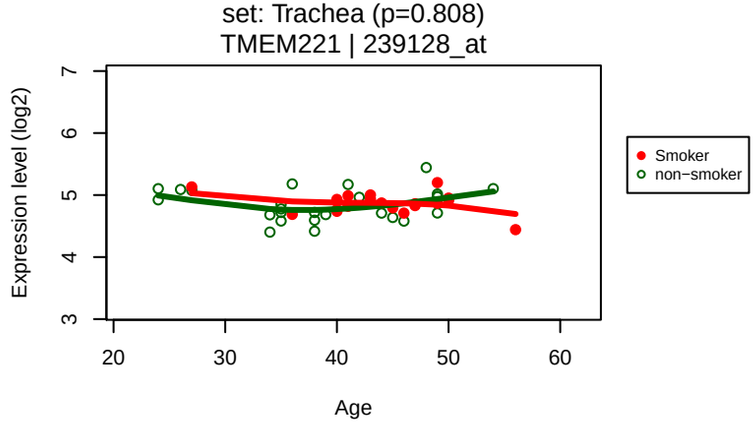

Gene expression ratio in Males
gene: TMEM221 | 239128_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3009
p=0.3623
p=0.2986
p=0.2
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM221 | 239128_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4502
p=0.7001
p=0.8584
p=0.7821
SmallAirway2
LargeAirway2
Trachea
Macrophage2

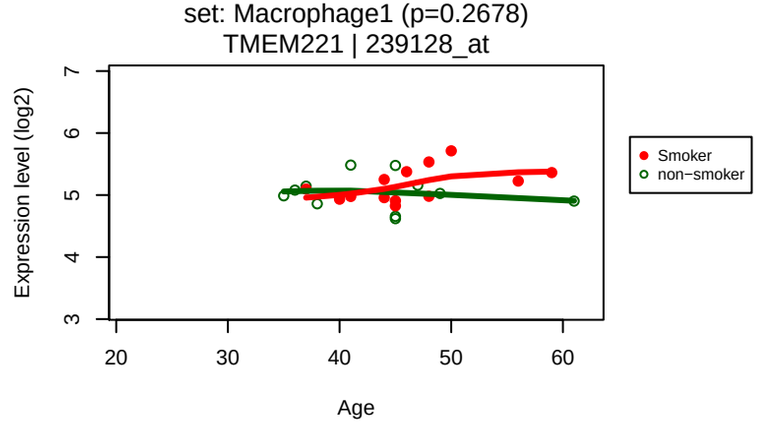

set: SmallAirway1 (p=0.0828)
TMEM221 | 239128_at
Expression level (log2)
Age
Smoker
non−smoker

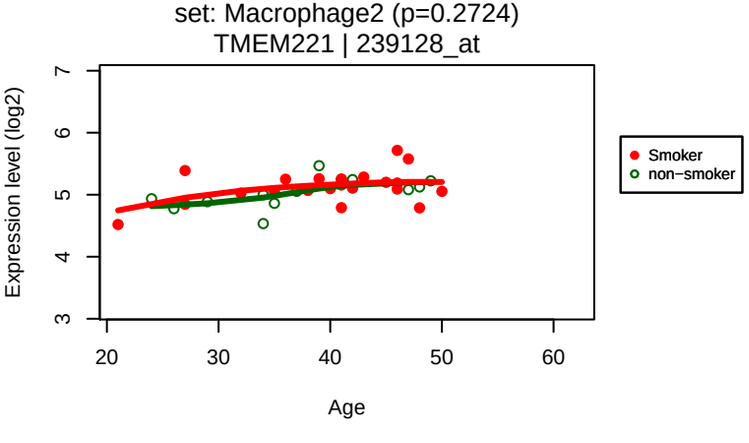

set: SmallAirway2 (p=0.124)
TMEM221 | 239128_at
Expression level (log2)
Age
Smoker
non−smoker

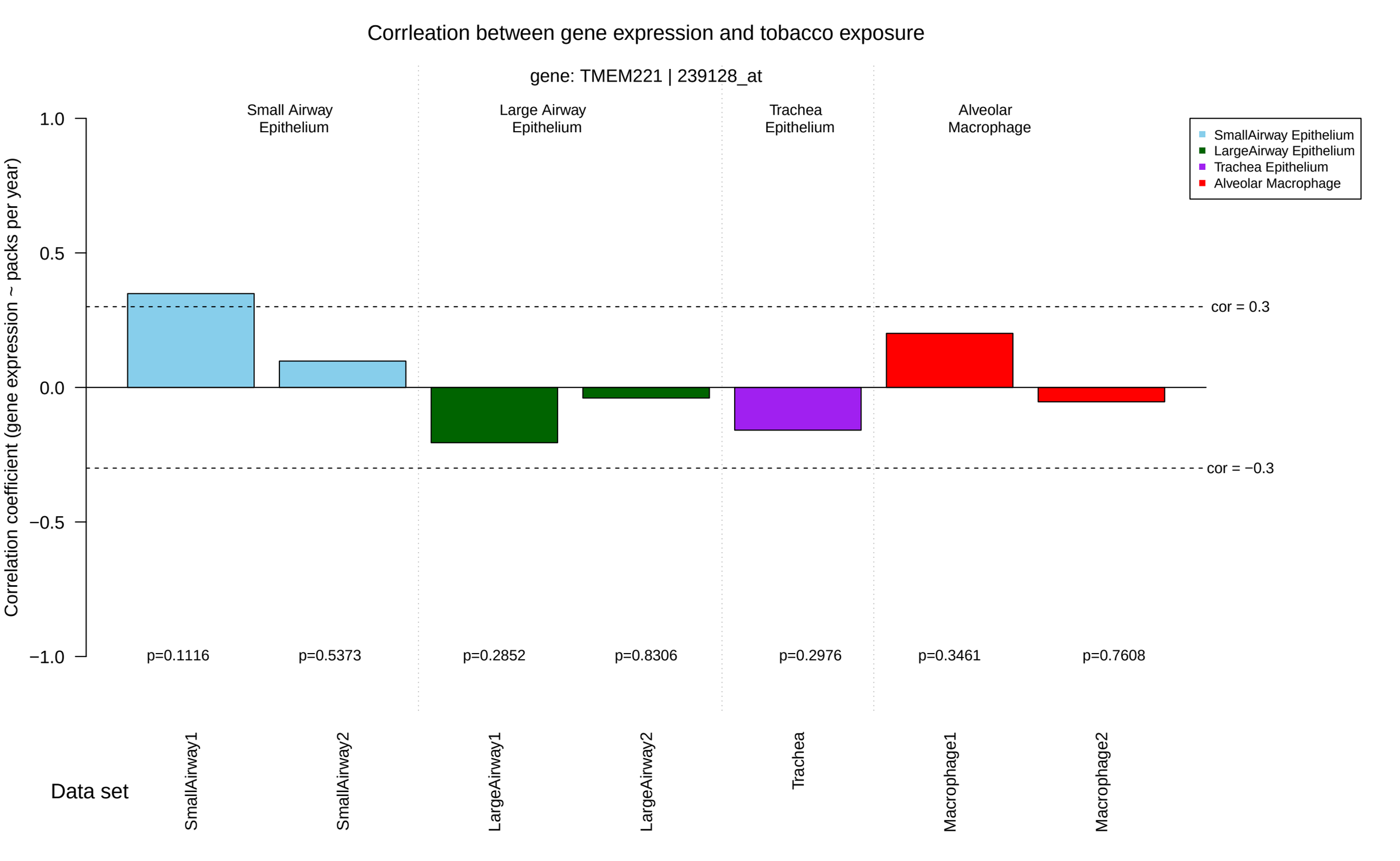

set: LargeAirway1 (p=0.672)
TMEM221 | 239128_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5537)
TMEM221 | 239128_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.808)
TMEM221 | 239128_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2678)
TMEM221 | 239128_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2724)
TMEM221 | 239128_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM221 | 239128_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1116
p=0.5373
p=0.2852
p=0.8306
p=0.2976
p=0.3461
p=0.7608
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2